

Junjie (Takku) Li

@ alvinbluy@gmail.com | 📁 Portfolio | 🔗 LinkedIn | 🐙 GitHub | 📍 Seattle (Open to Relocate)

EDUCATION

Northeastern University

Seattle, WA | Sep 2023 - Dec 2025

Master of Science in Computer Science (GPA: 3.9/4.0)

Teaching Assistant: 5004 Object Oriented Design, 5800 Algorithms

Nankai University

Tianjin, China

B.S. in Mathematics

EXPERIENCE

Berkshire Hathaway HomeServices: Intern

Seattle, WA | Jun 2024 - Aug 2024

Machine Learning Engineer

- Built data pipelines using **Airflow** to automate **ETL** processes, reducing data processing time by 17%.
- Customized and fine-tuned **deep learning models** (N-Beats series) using Pytorch for housing price and inventory predictions, improving model accuracy by 30%. Applied **Agile** principles to iteratively refine data acquisition processes and update models, resulting in a 25% reduction in development cycle time.

Temple University: PhD Research Assistant

Philadelphia, PA | Sep 2021 - May 2023

Design a tool to identified intriguing moments in a video for audiences.

- Preprocessed video and streaming comment data with **Pandas** and **Transformers**, leveraging **OpenCV** for frame extraction and **Sentence-BERT** for calculating real time synchronicity and intent similarity scores.
- Revealed more than 50% greater variance in content dynamics compared to topic modeling in after-consumption comments, providing deeper insights into how audiences engage with content.

PROJECTS

Search Engine for Emojis | [Link](#)

LLM-RAG Python/Go Web

"Emoji Search", a RAG-based AI search engine finding emojis based on web services.

- Adopted an event-driven architecture (**EDA**) for smoother search experience with four main components.
- Streamed user search queries through **Kafka** and integrated **Spark** to process large volumes of streaming data, improving real-time query handling by 20% and reducing batch processing time by 35% for analytics.
- Optimized microservice discovery and balanced queries, resulting in 30% fewer service failures and 15% faster response times at 5000 QPS. Containerized it in **Docker** and deployed it on **AWS EKS**.

AsyncTask Scheduler | Collaborated with [Hui Zhou](#) | [Link](#)

Distributed Go Framework

"Async4Ai", a light-weighted framework asynchronously processing interdependent tasks.

- Designed a distributed task-scheduling system that decouples producers from consumers, highlighting complete task management via web module with **RESTful API** for interaction between agents and database.
- Crafted loosely coupled table normalization & sharding logic in **MySQL** to improve database performance.
- Performed stress testing and improved performance from 500 QPS to 2000 QPS by designing an efficient caching mechanism with **Redis**, using distributed locks and reconfiguring the database connection pool.

Anonymous Academic Career Hub | [Link](#)

Full-stack JavaScript Web

"Business Job Market Rumors", a Blind-like discussion board for academic in business schools.

- Utilized **Express/Node** + **React** in a client-server model, integrating **Tailwind CSS**.
- Implemented role-based access control, context-based authentication using **JWT**, and automated content moderation using **Flask** to ensure a secure and respectful discussion environment.
- Deployed on **Azure App Service** via **GitHub Actions** with **MongoDB** hosted on **Azure Cosmos DB**.

TECHNICAL SKILLS

Languages & Databases: Python(5yrs), Java, Golang, C++, JavaScript, SQL; MongoDB, Redis

Frameworks: CUDA, Dask, PyTorch, Flask, Spring Boot, Express, Node, React, Angular, FastAPI

Tools: Maven, JUnit, Postman, GitHub Actions, RabbitMQ, Kubernetes, Docker, Spark

Miscellaneous: Linux, AWS([Certified Solutions Architect Associate](#)), Azure([Certified AI Engineer Associate](#))